

Conformations Search methods for molecular docking

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1 Basic concepts

Understanding molecular conformation the three-dimensional arrangement of atoms in a molecule is fundamental to computational chemistry. A molecule's conformation determines its chemical and biological properties, including reactivity, solubility, and binding affinity to biological targets. In the context of this book, this conformational search is performed withing the binding site of receptor, which aims to locate the bioactive conformation. Unlike global or local minimum conformation in solution, bioactive conformtion is one of the low energy conformations that shows most appropriate shape complementary with the binding site. This maximises favourable interactions between the ligand and receptor atoms. The success of structure-based drug design methods depends on identifying this bioactive conformation.

A typical small drug-like molecule may contain multiple rotatable bonds, each capable of adopting several energetically distinct rotameric states. With n rotatable bonds and approximately 3 accessible rotameric states per bond, the conformational space contains roughly 3^n distinct conformations. For a molecule with 12 rotatable bonds typical for many pharmaceutically relevant compounds, this yields nearly 530,000 possible conformations. Manual identification becomes impractical, necessitating automated conformational search methods.

1.1 Potential Energy Surface

The potential energy of a molecule is expressed as a function of the nuclear/atomic coordinates [?] given by:

$$V(\mathbf{R}) = V(R_1, R_2, \dots, R_{3N}), \quad (1)$$

where N is the number of atoms and $\mathbf{R}$ is the $3N$ -dimensional vector of Cartesian nuclear positions. For a non-linear molecule, the number of internal degrees of freedom is $3N - 6$, after removal of overall translation and rotation. Formally, a potential energy surface (PES) can be defined as the hypersurface obtained by plotting $V(\mathbf{R})$ against a chosen set of internal coordinates (bond lengths, angles and torsions). Such high-dimensional PES cannot be visualised for human interpretation, therefore, we must restrict plotting only a few important internal coordinates. These important internal coordinates are also known as collective variables that might be enough explain the behaviour of the entire chemical or biochemical system.

1.1.1 Stationary points and the Hessian matrix

While exploring the PES, there key parts of the PES that we are most interested in—locating the energy minimum. Mathematically, minimum or maximum points on the PES are located by differentiating the force field function describing the chemical system. In an ideal system, the minimum or maximum is defined as stationary points on the PES such that the first-order derivative of the function is zero and are classified by the eigenvalues of the Hessian matrix. The following equations illustrate this fact:

$$\frac{\partial V}{\partial q_i} = 0 \quad \text{for all internal coordinates } q_i, \quad (2)$$

$$H_{ij} = \frac{\partial^2 V}{\partial q_i \partial q_j}. \quad (3)$$

A local minimum has all positive eigenvalues (no imaginary frequencies), whereas a first-order saddle point (transition state) has exactly one negative eigenvalue (one imaginary frequency). The topology of the PES around these points controls conformational equilibria and reaction rates [?].

1.1.2 Energy profiles and reaction coordinates

In one dimension, for example rotation of a torsion angle, the PES reduces to an energy profile $V(s)$ along a reaction coordinate s . Energy minima correspond to stable conformers, while barriers may correspond to transition structures. Such barriers also suggest

1.2 Internal coordinates

For molecular mechanics representations, it is convenient to use internal coordinates such as:

- Bond stretches r ,
- Angle bends θ ,
- Improper torsions χ (for planarity/chirality),
- Proper torsions (dihedrals) ϕ ,
- Out-of-plane or improper torsion angles wherever necessary.

A generic functional form for the molecular mechanics potential energy is given by:

$$V(\mathbf{q}) = \sum_{\text{bonds}} k_r (r - r_0)^2 + \sum_{\text{angles}} k_\theta (\theta - \theta_0)^2 + \sum_{\text{torsions}} V_{\text{tor}}(\phi) + V_{\text{nonbonded}}, \quad (4)$$

where $V_{\text{nonbonded}}$ computes van der Waals and electrostatic interactions, and $V_{\text{tor}}(\phi)$ is the torsional potential.

1.3 Torsional potentials

Torsional terms are usually represented as a Fourier series,

$$V_{\text{tor}}(\phi) = \sum_n \frac{V_n}{2} [1 + \cos(n\phi - \gamma_n)], \quad (5)$$

where V_n are barrier heights, n the periodicity, and γ_n phase offsets. The barrier height controls the rate of interconversion between conformers. Very

low or negligible barrier makes the conformers indistinguishable, hence, considered equivalent for all practical purposes.

If we consider simple organic molecules:

- Ethane: $n = 3$, $\gamma = 0$, giving three equivalent staggered minima and three eclipsed maxima.
- Butane: interplay between $n = 3$ and $n = 1$ terms to distinguish anti- and gauche-staggered minima.

1.4 Rotatable bonds

Chemically, a rotatable bond can be defined as follows:

- A single (non-aromatic) bond (σ bond) between heavy atoms
- That is not in a ring
- And is not terminal (i.e. not $X-CH_3$ or similar simple substituents)
- Excluding amide C-N bonds, and other formally conjugated single bonds with high torsional barriers.

1.5 Conformational ensembles and thermodynamics

Given a set of conformers i with relative energies E_i , the Boltzmann-weighted population at temperature T is

$$p_i = \frac{\exp(-E_i/k_B T)}{\sum_j \exp(-E_j/k_B T)}, \quad (6)$$

so that low-energy conformers dominate the bulk properties. Computational workflows therefore:

- Sample a sufficiently diverse ensemble on the PES (system- and method-dependent).
- Reweight or rescore conformers with higher-level methods when needed.
- Consider entropic contributions associated with loss of conformational freedom upon binding.

2 Metrics to ensure conformational sampling

2.1 Energy-Based Metrics

Key metrics characterize conformational search results:

$$\Delta E = E_{\text{conformer}} - E_{\text{global minimum}} \quad (7)$$

The relative energy ΔE expresses each conformation’s energy relative to the lowest-energy structure. The Boltzmann population at temperature T predicts the probability of observing a particular conformation:

$$P_i = \frac{e^{-\Delta E_i/k_B T}}{\sum_j e^{-\Delta E_j/k_B T}} \quad (8)$$

where k_B is Boltzmann’s constant ($1.987 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$). At 298 K, conformations within 2 kcal/mol of the global minimum constitute $> 10\%$ of the population and should be included in conformational ensembles.

The RMSD quantifies geometric similarity between two conformations:

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N d_i^2} \quad (9)$$

where d_i represents the distance between atoms i after optimal superposition. RMSD thresholds (typically 1.0–2.0 Å) filter redundant conformations in search output.

3 Conformational search methods

One of the most challenging aspect of pose prediction is the conformational search algorithm. In general, there are two main classes of conformational search methods, systematic and stochastic. The users do not always have a choice to select the search algorithm because the developers usually select based on their expertise and code development. Nevertheless, there are programs, such as AutoDock, that allow users to select the search algorithm.

3.1 Systematic Methods

The systematic search algorithm rotates all possible rotatable bonds (single bonds in the ring structures cannot be rotated as freely as open-chain structures) systematically by a fixed interval to explore all possible conformations. Such a method can explore all possible conformations, however, the complexity increases exponentially as the number of rotatable bonds increases. There are pruning algorithms that act as a “bump check” that rule out torsion angles that lead to significant atomic overlaps. The deterministic nature of systematic sampling guarantees that the same input will yield the same set of conformers, which is advantageous for benchmarking and reproducibility. However, the combinatorial explosion of grid points with increasing rotatable bonds quickly becomes prohibitive, especially for highly flexible ligands, motivating the use of hybrid schemes or coarse-grained grids in practice [?, ?].

3.1.1 Grid-based enumeration

The simplest systematic approach divides torsion angle space into discrete intervals (e.g., 10° or 30°) and evaluates all possible combinations. For a molecule with n rotatable bonds and 36 intervals per 360° rotation:

$$N_{\text{conformations}} = 36^n \quad (10)$$

This method guarantees finding the global minimum but becomes computationally prohibitive for large n (combinatorial explosion). Grid enumeration works well for rigid molecules with 2–4 rotatable bonds but is impractical for flexible compounds.

In the context of molecular docking, the search space is limited to the binding pocket, often provided as a grid. Therefore, the systematic conformational search engine operates by discretizing the relevant degrees of freedom typically torsional angles on a this grid. It then generated and evaluates all combinations of torsion angles within that grid. For a molecule with n rotatable bonds, each torsion is sampled at a fixed interval, producing a Cartesian product of values $\phi_1 \in \phi_1^{(1)}, \dots, \phi_1^{(k)}, \phi_2 \in \phi_2^{(1)}, \dots, \phi_2^{(k)}$ and so on. At each grid point, the geometry is energy-minimized, and the resulting conformers are clustered and ranked by energy or empirical scoring functions. Because the search is exhaustive within the chosen resolution, systematic methods can in principle locate all sterically allowed conformations

of a given molecule, provided the grid is sufficiently fine and the force field adequately describes the intramolecular energetics [?].

3.1.2 The Z method

The Z method [?] is a variant of grid-based conformational search protocols that employs statistical information on the rotamers. It partitions the conformational space into subspaces and then applies repeated, uniform changes to a common starting structure, generating large ensembles of trial conformations in parallel. This approach allows rigid systematic searches in high dimensional spaces by combining grid-based sampling with statistical information from rotamer libraries. The Z Method can also be used hierarchically: low resolution searches identify approximate regions of interest, which are then refined with finer grids or energy minimization protocols. The key advantage is the ability to couple systematic search with probabilistic rotamer preferences, improving the sampling efficiency while preserving the exhaustiveness of the underlying grid. However, to the best of our knowledge, this method has only been applied protein structure refinement. We did not find any literature references indicating its use as conformational search tool for small molecules or in molecular docking.

3.2 Fragmentation method

In this method, the molecule is broken down into individual fragments. The fragments are then docked in the most suitable sub-pocket in the binding site. The entire molecule is then sequentially built by adding the linkers. This follows a systematic search for the linker conformation that optimally holds the fragments in their respective sub-pockets. The fragmentation method selects the rigid molecular components, for example ring structures, as fragments, and the flexible components as linkers. In doing so, this method reduces the computational complexity compared to the systematic search by focusing on small components, more particularly the linkers. FlexX and DOCK are examples of the docking algorithms that employ this method as its conformational search algorithm.

3.3 Stochastic Methods

Stochastic techniques utilize random sampling and probabilistic methods to explore the conformational space. They work by randomly making changes, preferably, to the rotatable bonds and evaluates the energy following the change. Metropolis Monte Carlo, and Genetic algorithm are popular stochastic search methods that are used as conformational search tools in docking programs.

3.3.1 Monte Carlo

It works on the principle of random sampling for a predetermined number of iterations to locate optimum conformation. The search algorithm starts by making a random changes in any of randomly selected rotatable bond. Following which, it evaluates the energy and compares it with the starting conformation. If the energy of the new state is lower than the previous state, then the conformation is accepted. If the energy of the new state is higher than the previous state, then it assigns a random number between 0 and 1 and evaluates its probability against the Boltzmann distribution; if the random number assigned is greater than the probability, then the new state is accepted otherwise rejected. In case of the rejection, the algorithm restarts from the previous conformations. Glide docking program is also known to employ Monte Carlo simulation in its docking algorithm to improve its pose prediction accuracy.

3.3.2 Genetic Algorithm (GA)

This method is motivated from the process of natural selection, that evaluates energy or docking score as the fitness criteria for selection and breeding. Akin to natural selection, the conformational degrees of freedom are encoded as binary numbers which represents specify values of the torsion angles. At the beginning, several random mutations are made in the binary numbers (interchanging “0” to “1”) and generating an initial population. The fitness score is evaluated for each member. Following which the top ranked (fittest members) are retained for breeding. To generate future generations, either cross-over is performed or mutations. This process is repeated for a predetermined number of iterations. Autodock and GOLD are examples of docking programs that use genetic algorithm as their conformational search tool.

3.4 Molecular Dynamics Simulations

At this point, it is appropriate to gently introduce another powerful conformational search method, namely *molecular dynamics (MD) simulations*. It simulates the atomic motions as a function of time by solving the Newton's equation of motion. MD simulations and molecular docking are complementary that are commonly employed in structure-based drug discovery to understand the protein-ligand interactions and their dynamics aspects. For computational efficiency, molecular docking algorithms treat receptors atoms rigid and the ligand as flexible. This leads to incorrect pose prediction when induced fit binding is observed. Therefore, MD simulation is an elegant method to incorporate such induced fit effects. This can be achieved in two ways, as shown below:

- Pre-docking step to sample various conformations of the receptor without the influence of the ligand. This can serve as multiple receptor conformations for docking.
- Post-docking step to refine the docked receptor-ligand conformation by allowing the complex to evolve to their realistic conformation, which would be more probable physiologically.

3.5 Simulated Annealing

Simulated annealing (SA) is a concept more prevalent in engineering science, where malleability in metals/alloys is employed to manufacture cables for a vast variety of applications. For example, a long steel cable can be manufactured from a block of steel by heating then beating and then cooling. This cycle is repeated until the desired shape, size and length is achieved, which dictates the tensile strength of the steel cable. This heating-beating-cooling-heating cycle is simulated annealing. The heating increases the softness of the steel by making the atoms move apart by gaining kinetic energy at high temperatures. Thus, making the beating process easier and preventing any fracture or permanent deformation. The cooling stage traps the atoms in the beaten state. At the end of each cycle, the shape of the steel block progressively approaches the desired shape and size without losing the inherent properties of steel.

Previously, discussed idea was implemented as a powerful stochastic optimization technique widely used in conformational search for molecular dock-

ing, where the goal is to locate low-energy ligand and protein conformations compatible with experimental binding modes. The heating in this case, simply meant adding more velocity to the atoms so that they can surmount the nearest energy barrier. Similarly, the cooling step meant reducing the velocity such the system now falls in the nearest local minimum. The heating and cooling is carried out iterative to ensure maximum sampling of the PES. This mapping of thermodynamic annealing onto conformational space makes SA particularly suited to global-optimization problems such as flexible-ligand docking, where the search space is rugged and high-dimensional.

1. *Heating phase*: Starting from an initial conformation, the system is rapidly heated to high temperature (typically 800–1000 K), providing energy to overcome conformational barriers.
2. *Annealing phase*: Temperature is gradually reduced according to a cooling schedule, allowing the system to explore broader conformational space while progressively favoring lower-energy structures.
3. *Minimization phase*: At low temperature (50–100 K), geometry optimization drives the system to a nearby local minimum.

The SA protocol may be repeated 100+ times from different initial conformations, generating diverse low-energy structures. Effectiveness depends critically on cooling rate: too rapid cooling traps the system in local minima; too slow cooling demands excessive computation.

3.6 Distance Geometry Methods

3.7 Hybrid systematic–stochastic schemes

4 Practical Considerations

4.1 Conformational flexibility in acyclic systems

For simple alkanes, conformational preferences are well rationalised by *Newman projections*:

- Staggered conformers minimise torsional strain and often benefit from hyperconjugative stabilisation.

- Eclipsed conformers are higher in energy due to torsional and sometimes steric/electrostatic repulsions.

In butane, the anti-staggered conformer is lowest in energy, with gauche conformers slightly higher but still populated at room temperature; this basic model generalises to many drug-like substituents.

Hyperconjugation and stereoelectronic effects (e.g. the gauche effect, anomeric effect) often bias torsional preferences away from purely steric expectations. These effects are naturally captured in quantum chemical PES scans and, when parametrised appropriately, in modern force fields.

4.2 Conformational flexibility in ring systems

4.2.1 Ring strain components

Cycloalkanes illustrate several types of strain:

- *Angle strain*: deviation of bond angles from the ideal tetrahedral value ($\approx 109.5^\circ$).
- *Torsional strain*: eclipsing interactions between neighbouring bonds.
- *Steric (non-bonded) strain*: 1,3-diaxial interactions, transannular repulsions in medium/large rings, etc.

Small rings (cyclopropane, cyclobutane) are strongly angle- and torsionally strained; cyclopentane relieves torsional strain by puckering; cyclohexane adopts near strain-free conformations.

4.2.2 Cyclohexane: chair, boat and twist-boat

In cyclohexane, the most important conformations are:

- Two equivalent chair conformers, related by a ring flip.
- Higher-energy boat and twist-boat conformations.

In the ideal chair, all C–C bonds are staggered and the internal bond angles are close to tetrahedral. A ring flip exchanges axial and equatorial positions; for monosubstituted cyclohexanes, the equilibrium strongly favours the equatorial conformer for bulky substituents due to reduced 1,3-diaxial repulsions.

From a computational perspective, the PES in the relevant ring-puckering coordinates shows:

- Two deep minima (chairs),
- A pathway via twist-boat and boat conformations connecting them,
- Barriers that set the timescale for ring inversions.

4.2.3 Conformational analysis of substituted and heterocyclic rings

Substitution and heteroatoms modify ring conformations and relative energies:

- Electronegative or σ -accepting substituents can introduce stereoelectronic effects (e.g. anomeric effect in pyranoses).
- Heteroatoms change bond lengths/angles and steric profiles, altering A-values (axial penalty) and flipping preferences.
- Fused ring systems transmit conformational bias conformational transmission, important in polycyclic scaffolds used in medicinal chemistry.

Accurate treatment of these systems often requires a combination of high-quality force fields and validation against quantum chemical PES scans or experiment.

4.3 Accounting for solvation effects

Bioactive conformations depend on the solvent environment. Implicit solvation models (Generalized Born, Poisson-Boltzmann) add solvation energy terms, affecting the conformational landscape relative to gas-phase calculations.

4.4 Clustering of Results

Why we need clustering? and how does this ensure use diverse conformations for next step.

RMSD:

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_i |\mathbf{r}_i - \mathbf{r}'_i|^2}$$

Clusters identify recurring poses.

4.5 Validation Metrics

Conformational search quality is evaluated by:

1. *Recovery of Crystallographic Conformations*: For validation datasets with known bioactive structures (e.g., protein-bound ligand conformations), the minimum RMSD from the ensemble to experimental structure indicates success. Recovery within 2 Å RMSD is considered good.
2. *Conformational Diversity*: The range of geometries in the final ensemble should comprehensively represent the conformational space. Metrics include distribution of dihedral angles and range of relative energies.
3. *Computational Efficiency*: Wall-clock time and peak memory usage relative to molecular size determine practical viability. For drug discovery pipelines screening millions of compounds, efficiency is paramount.

5 Summary

Conformational search is the computational engine that enables docking. Understanding both the physics of the scoring function and the algorithms used to explore conformational space is essential for reliable predictions. Hybrid strategies that combine stochastic global exploration with local refinement currently offer the best trade-off between efficiency and accuracy.